# Week 5 — Activity: Isostatic Correction

Geophysics 3A: Potential Fields & Geothermics

March 27, 2026

## Objective

In this problem, you will derive the isostatic gravity correction under the assumption of Airy isostasy, starting from a Bouguer slab formulation and progressing to a practical implementation using topography and a reference crustal thickness.

### Background

Consider a reference Earth model with a constant crustal thickness $T_0$. Surface topography $h(x)$ is compensated by a crustal root $r(x)$ such that columns of equal width have equal mass above a compensation depth.

Let:

- $\rho_c$ = crustal density
- $\rho_m$ = mantle density
- $h(x)$ = surface elevation
- $r(x)$ = additional root thickness relative to $T_0$

### Geometry

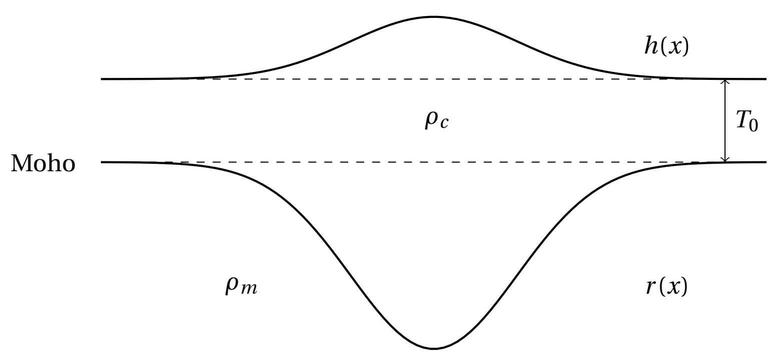


Figure 1: A mountain with a topographic expression $h(x)$ and compensating root $r(x)$.

### Part A: Airy Isostasy

1. Starting from mass balance between a reference column and a column with topography, show that:

$$\rho_c h = (\rho_m - \rho_c) r$$

2. Solve for the root thickness $r(x)$ in terms of $h(x)$.

## Part B: Gravity of an Infinite Slab

The gravitational attraction of an infinite horizontal slab of thickness $t$ and density $\rho$ is:

$$g = 2\pi G\rho t$$

3. Write an expression for the gravitational effect of the topography.
4. Write an expression for the gravitational effect of the crustal root. Carefully define the density contrast.

## Part C: Substitution and Simplification

5. Substitute your expression for $r(x)$ into the root gravity term.
6. Show that the total gravity effect of the topography plus its Airy root simplifies to:

$$g_{\text{total}} = 0$$

## Part D: Practical Isostatic Correction

In practice, crustal thickness is not known everywhere. Instead, we assume a reference thickness $T_0$ and compute $r(x)$ from observed topography.

7. Explain why the isostatic correction is model-dependent.
8. Write the final expression for the isostatic correction in terms of $h(x)$ only.

## Target Result

You should show that the gravity effect of the compensating root is:

$$g_{\text{root}} = -2\pi G\rho_c h$$

and that, under perfect Airy compensation, the combined effect of topography and root cancels exactly.

## Extension (Optional)

Discuss how this result would change if:

- compensation were incomplete
- the lithosphere behaved elastically (flexural isostasy)